

CA-Software Assignment - 3

Trends of cloud services Overview - write a three-paragraph report covering: 1) the benefits of serverless architecture, 2) the concept of progressive web apps (PWAs), & 3) the role of AI & Machine learning in software architecture. Then, in one paragraph, describe the cloud computing service models (SaaS, PaaS, IaaS) & their use cases.

1. Benefits of Serverless Architecture:

- No server management: Automatically scales with demand.
- Cost-efficient: Pay only for actual usage.
- Faster development: Developers focus only on business logic.
- Use case: Running functions on AWS Lambda when an order is placed in an e-commerce app.

2. Progressive Web Apps (PWAs):

- Definition: Web applications that behave like native apps on mobile.

• Key features:

- Offline access with service workers
- Push notifications
- Installable to home screen
- Use Case: shopping site that works without networks, stores cart locally, & syncs later.

3. Role of AI & ML:

- Personalization: Product recommendations based on User behavior.
- Chatbots: AI-powered assistants for customer support.
- Demand Forecasting: Predict inventory needs using ML Models.
- Fraud detection: Analyze patterns in transactions.

Cloud Service Models & Use Cases:

- SaaS (Software as a Service):
 - End-User applications like Gmail, Dropbox
 - Use-Case: Shopify for managing e-commerce.
- PaaS (Platform as a Service):
 - Developer tools to build & deploy apps

- Use Case: Google App Engine to host web apps
- IaaS (Infrastructure as a Service):
 - Virtual machines, storage, networking
 - Use Case: AWS EC2 for hosting servers.